



Kishoreganj University
3rd Year 1st Semester B. Sc. (Engg.) Final Examination-2024
Department of Computer Science and Engineering
CSE 3107: Information System Design (3 Credits)

Time: 3 Hours

Full Marks: 70

Figures shown in the right margin indicate full marks.

Answer 5 out of 7 questions.

- ✓ 1. Suppose you are part of a student development team at Kishoreganj University that has been assigned to design and develop a system called Smart Campus Connect. The goal of this system is to make campus life more connected and convenient by allowing students to discover and register for university events, manage club memberships, book rooms for activities, and receive event announcements in one place. Your team will follow the complete software design and development process, from gathering requirements and creating system designs to managing the project using professional tools and best practices.
- a. Identify the key stakeholders involved in the Smart Campus Connect project and describe the type of information you would gather from each group. 4
- b. Explain how you would plan and conduct the requirement collection process for developing the Smart Campus Connect system. Discuss how you would select appropriate elicitation techniques and ensure the collected requirements are accurate, complete, and free from bias. 5
- c. After collecting the initial requirements, your team needs to prepare a Software Requirements Specification (SRS) document. Explain what major sections you would include in the SRS and why each section is important. 5
2. a. Explain the Waterfall model in software development. Considering the nature of the Smart Campus Connect project described in Question 1, discuss whether this model would be suitable and justify your answer based on its characteristics and limitations. 4
- b. The team decides to adopt a Prototyping approach instead. Explain how this approach works, why it might be more appropriate for developing the Smart Campus Connect system, and what advantages it offers in terms of stakeholder engagement and requirement refinement. 5
- c. Describe the key phases of the V-Model and explain how verification and validation are embedded at each stage. In your explanation, clarify how the V-Model ensures both process discipline and early defect detection. 5
3. a. Develop a Use Case Diagram for the Smart Campus Connect system involving the actors Student, Event Organizer, and Admin. 4
- b. Draw or describe a Level-0 Data Flow Diagram (DFD) that represents how event registration information moves through the Smart Campus Connect system. In your answer, explain how the DFD supports understanding of the system's logical flow and data dependencies. 5
- c. Explain how a Class Diagram and a State Diagram, when used together, provide complementary perspectives on a system. Discuss what each reveals about the design of Smart Campus Connect and why both are essential for complete system modeling. 5

- 4 a. In the Smart Campus Connect prototype, some students said they could not understand what the navigation icons meant. What will you do in this situation to solve the problem as a designer? 4
- b. During user testing, students mentioned that the home screen looked too crowded, and the team now wants to decide how to improve the interface before the next round of testing. As a designer, what specific changes would you suggest to make the layout clearer and more user-friendly? 5
- c. The design team is deciding how to collect user feedback for the next prototype. They can either ask users to rate each screen or observe them performing real tasks. Which method would you choose, and why? 5
- 5 a. There is a common printing service for printing secured official documents in an office which is used by different departments. Sometimes, multiple departments send lots of printing request simultaneously and want to get assured of their request is being addressed. What technology should be recommended by the designer of such a system? Also draw a block diagram for the required design. 5
- b. You have developed an application for different clients. You need to deploy that quickly in different types of servers of the clients. What DevOps technology will serve your purpose? 5
- c. In your hardware infrastructure, servers of different capacity are being used. Which load balancing algorithm is best suited for your case? 4
- ✓ 6 a. Explain why Version Control Systems (VCS) such as Git are important when several students work together on a project like Smart Campus Connect. 4
- f b. Your team members are working on different parts of the Smart Campus Connect system. They are unsure whether to work directly on the main branch or to create separate branches for each feature. What would you suggest they do, and why? 5
- c. While merging code, both Badhon and Pronomita edited the same file named eventController.js, which caused a merge conflict. As part of the development team, what steps would you take to fix the conflict properly so that both developers' work is included? 5
- 7 a. As project manager how do you form your team for the following software projects: 4
 i) A project for solving complex scientific problem
 ii) Development of ERP with many modules
 iii) A software which must be ready at the beginning of the next year
- b. Define software failure. What are the components of failure cost? How can software failures be prevented during development? 3
- c. Explain the term measure and metric with examples in the discipline of software measurement. Also explain two measurement principles with necessary examples. 4
- d. Your goal is to evaluate the performance of the programmers. Define the metrics to achieve this goal using goal oriented software measurement. 3